

ASSIGNMENT3:(30%)

Firebase

Strengthen your ability to apply your knowledge of Firebase

Overview

Overview: Building off the weather app by adding some customization and authentication to your application

The user will have the ability to register and login into their weather app. The user will have ability to save different weather locations.

Users will have ability to also remove weather locations from their app.

This assignment is a continuation from the previous weather assignment.

Your application will leverage weatherapi.com site to retrieve weather data.

Your app must be written in either Java or Kotlin. Hybrid platforms (e.g., React Native) are not acceptable.

You can work in a **group of Three** from your section.

Only use tech stack that we have used in class. Any unauthorized technologies will result in an automatic zero.

Any compressed project over 50MB will receive a mark of zero.

Instructions:

Part 1 Weather Search Screen(20 Marks):

1. **Mandatory:** Ability to search weather for a specific region.
2. Use weatherapi.com to search by city name.
3. A list of weather locations should dynamically appear based on the city name provided.
4. The UI should give the user the ability to save a city name to their favorite city locations.

Part 2 Home Page and Delete Locations (10 Marks):

1. **Mandatory:** After the user has finished authentication, the app will direct the user to the home page.
2. The home page will list all the saved locations, provided by firebase firestore.
3. Users should have the ability to remove regions from the home page

Part 3 Navigation Menu (10 Marks):

1. **Mandatory:** At the bottom of the page should contain two menu options, Search and Favorites
2. Search menu option navigates user to search page. Favorite menu option navigates users to the favorites page.

Part 4 Login/Register Screen (10 Marks):

1. **Mandatory:** Ensure your Login Screen's logic (Activity) connects and works with Firebase Authentication (5 Marks).
2. Your Registration Screen should include UI Elements (Text Fields) that allow the user to login as a returning user (5 Marks).
3. Your Login Screen should include a UI Element that allows the user to Login (2.5 Marks)
4. If the Login attempt is unsuccessful, an alert (or other UI Elements) should be displayed that informs the user that their Login attempt has failed. (2.5 Marks).

Part 5 Your App is Aesthetically Pleasing (20 Marks):

1. You've designed your App with good Aesthetic standards including appropriate margins, padding, colors, contrast and overall layout while always keeping the user in mind

Part 6 Code Quality: (10 Marks)

1. Ensure your app includes the following components (**10 Marks**).
 - a. Clean Build
 - b. MVVM Architecture.
 - c. LiveData.
 - d. ViewBinding.
 - e. WeatherAPI.
 - f. Recycler View
 - g. OKHTTP

h. Firebase.

Part 7 Video Presentation: (20 Marks)

1. Create a Short Video presentation on YouTube or another streaming provider. You must include a short PowerPoint (or Google Slides) Slide Deck that includes a single slide to start your video
 - a. The **first** (and only) **Slide** of your Slide Deck must include a **current image** of you and team members (no avatars allowed) that is displayed appropriately on the page. You must also include your Full Name(s), Student ID(s), the Course Code, Course Name, and your Assignment information.
 - b. You will **demonstrate** your app's functionality.
 - c. You must **explain how your apps code works**.
 - d. Sound for your Video must at an appropriate level so that your voice may be **clearly heard**.
 - e. Your screen resolution should be set so that your code and app demonstration is **clearly visible**.
 - f. Your Short Video should run no more than 10 minutes.
 - g. Your Video may be unlisted, but must be accessible to the instructor
 - a. Make sure the instructor has permission to view the video.

Submission Details:

Your submission should include:

- a. A **zip archive** of your Project files (Mandatory)
- b. Format for Zip file: **assignment{Number}_{StudentLastName}.zip**
- c. **Submit your PPT file** which has a link embedded **to your Video Presentation**

Assessment:

You will be assessed on the following:

	Not Quite	Good Work	Awesome
Weather Search (20 Marks)	(0 - 8 marks)	(9 - 17 marks)	(18 - 25 marks)
Home Page and Delete Locations (10 Marks)	(0 - 5 marks)	(6 -10 marks)	(11 - 15 marks)
Navigation Menu (10 marks)	(0 - 5 marks)	(6 -10 marks)	(11 - 15 marks)
Login/Register Screen (10 marks)	(0 - 5 marks)	(6 -10 marks)	(11 - 15 marks)

	Not Quite	Good Work	Awesome
Your App is Aesthetically Pleasing (20 Marks)	(0 - 3 marks)	(4 - 7 marks)	(8 - 10 marks)
Code Quality MVVM LiveData Firebase (10 marks)	(0 - 3 marks)	(4 - 7 marks)	(8 - 10 marks)
Video Presentation Your short video must demonstrate your site and describe your code (20 marks)	(0 - 3 marks)	(4 - 7 marks)	(8 - 10 marks)

Due Date:

Check Class Website

Late Submission:

1 day late	-20% cumulative penalty
2 days late	-40% cumulative penalty
3 days late	-100% cumulative penalty

Assignment Weight:

Assignment Three is worth 30% of your final grade.

Please submit your assignment online under Assignments > Assignment Three.

!important

Please ensure that any work you submit is your own unique and independent work. Work submitted that is found to be not your own unique, and independent work will be subjected to a grade of 0 and considered to be academic misconduct.

Login

Login

Register

Favorites

Weather 1 -

Weather 2 -

Weather 3 -

Favorites

Search

Favorites Details

Weather Image

Current Temp

Hi/Low

Remove

Back

Register

Register

Search

Toronto

Search

Location 1 +

Location 2 +

Favorites

Search